Experiment 7

Aim: Perform word sense disambiguation using WordNet

Theory:

Word Sense Disambiguation (WSD) is the process of identifying the correct meaning of a word based on its context. Many words in natural language have multiple meanings, and WSD helps in selecting the most appropriate sense. For example, the word "bank" can refer to a financial institution or the side of a river depending on the sentence.

WordNet is a lexical database that groups words into sets of synonyms called synsets. Each synset represents a different meaning of a word and provides definitions, examples, and relationships such as synonyms, antonyms, hypernyms, and hyponyms. These features help in determining the correct sense of a word.

In Word Sense Disambiguation using WordNet, the ambiguous word is first identified in a sentence. Then all possible meanings (synsets) of the word are retrieved from WordNet. The context words in the sentence are compared with the definitions and examples of each synset. The sense that best matches the context is selected as the correct meaning.

This method is useful in many NLP applications such as machine translation, information retrieval, text understanding, and question answering, where identifying the correct meaning of words is important.

Source Code and Outputs:

```
!pip install nltk

import nltk
from nltk.corpus import wordnet as wn
from nltk.wsd import lesk
from nltk.tokenize import word_tokenize
```

```
Requirement already satisfied: nltk in /usr/local/lib/python3.12/dist-packages (3.9.1)
Requirement already satisfied: click in /usr/local/lib/python3.12/dist-packages (from nltk) (8.3.1)
Requirement already satisfied: joblib in /usr/local/lib/python3.12/dist-packages (from nltk) (1.5.3)
Requirement already satisfied: regex>=2021.8.3 in /usr/local/lib/python3.12/dist-packages (from nltk) (2025.11.3)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from nltk) (4.67.3)
```

```
nltk.download('punkt')
nltk.download('punkt_tab')
nltk.download('wordnet')
nltk.download('omw-1.4')
```

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Package punkt is already up-to-date!
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt_tab.zip.
[nltk_data] Downloading package wordnet to /root/nltk_data...
[nltk_data]   Package wordnet is already up-to-date!
[nltk_data] Downloading package omw-1.4 to /root/nltk_data...
[nltk_data]   Package omw-1.4 is already up-to-date!
True
```

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[nltk_data]   Package wordnet is already up-to-date!
[nltk_data] Downloading package omw-1.4 to /root/nltk_data...
[nltk_data]   Package omw-1.4 is already up-to-date!
True
```

```
sentence = "I went to the bank to deposit money"
words = word_tokenize(sentence)

print(words)
```

```
['I', 'went', 'to', 'the', 'bank', 'to', 'deposit', 'money']
```

```
target_word = "bank"

sense = lesk(words, target_word)

print("Word:", target_word)
print("Sense:", sense)
print("Definition:", sense.definition())
print("Examples:", sense.examples())
```

```
Word: bank
Sense: Synset('savings_bank.n.02')
Definition: a container (usually with a slot in the top) for keeping money at home
Examples: ['the coin bank was empty']
```

```
sentence2 = "The river bank was full of trees"
words2 = word_tokenize(sentence2)

sense2 = lesk(words2, "bank")

print("Sentence:", sentence2)
print("Sense:", sense2)
print("Definition:", sense2.definition())
```

```
Sentence: The river bank was full of trees
Sense: Synset('deposit.v.02')
Definition: put into a bank account
```

```
word = "bank"

synsets = wn.synsets(word)

for i, syn in enumerate(synsets):
    print(f"Sense {i+1}: {syn.name()}")
    print("Definition:", syn.definition())
    print("Examples:", syn.examples())
    print()
```

```
Sense 1: bank.n.01
Definition: sloping land (especially the slope beside a body of water)
Examples: ['they pulled the canoe up on the bank', 'he sat on the bank of the river and watched the currents']

Sense 2: depository_financial_institution.n.01
Definition: a financial institution that accepts deposits and channels the money into lending activities
Examples: ['he cashed a check at the bank', 'that bank holds the mortgage on my home']

Sense 3: bank.n.03
Definition: a long ridge or pile
Examples: ['a huge bank of earth']

Sense 4: bank.n.04
Definition: an arrangement of similar objects in a row or in tiers
Examples: ['he operated a bank of switches']

Sense 5: bank.n.05
Definition: a supply or stock held in reserve for future use (especially in emergencies)
Examples: []

Sense 6: bank.n.06
Definition: the funds held by a gambling house or the dealer in some gambling games
Examples: ['he tried to break the bank at Monte Carlo']
```

Learning Outcomes:

I am able to understand and apply WordNet for word sense disambiguation by analyzing contextual relationships between words and selecting the most appropriate meaning of ambiguous terms, which enhances the semantic understanding and accuracy of Natural Language Processing systems.